

Automated Research Discovery

arxiv-sanity, TF-IDF Search & LLM Paper Agents

Andrej Karpathy | arxiv-sanity.com | [arxiv-sanity-lite](https://arxiv-sanity-lite.com)

Introduction

Andrej Karpathy developed [arxiv-sanity](https://arxiv-sanity.com) (arxiv-sanity.com) and [arxiv-sanity-lite](https://arxiv-sanity-lite.com) as tools to tame the flood of machine learning papers on arxiv.org. These tools implement semantic similarity search, personalized recommendations, and paper discovery at scale. This document covers the technical design and broader implications for automated research discovery.

1. The Problem: arxiv Overload

The cs.AI, cs.LG, and cs.CL categories on arxiv together receive 400–600+ new papers per day. No researcher can manually screen this volume. Automated tools must filter, rank, and surface relevant papers based on personal interest models.

2. arxiv-sanity-lite: Architecture

Karpathy's redesigned [arxiv-sanity-lite](https://arxiv-sanity-lite.com) (2022+) uses a lean, interpretable stack:

Feature Extraction: TF-IDF vectors over paper abstracts (50K-token vocabulary). Simple but highly effective for research paper similarity.

Similarity Model: SVM classifier with RBF kernel trained on papers the user has 'liked'. Scores all new papers in the corpus relative to user preferences.

Retrieval: Matrix multiply over normalized TF-IDF features for semantic similarity. No neural embeddings needed — sparse TF-IDF achieves strong recall on arXiv.

Recommendation: Time-decayed scoring: new papers in liked topics surface within hours.

Interface: Flask + SQLite backend; minimal frontend. ~500 lines of Python total.

3. TF-IDF for Paper Search

The TF-IDF score for term t in document d :

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) * \log(N / \text{DF}(t))$$

where TF = term frequency in document, N = total documents, DF = documents containing term. Papers sharing rare technical terms (e.g., 'RLHF', 'Constitutional AI', 'chain-of-thought') score highly even without semantic embeddings.

4. LLM Agents for Research Discovery

Beyond Karpathy's tools, modern approaches use LLM agents to automate literature search:

Perplexity AI: LLM + real-time web search. Cites sources; good for recent work.

Semantic Scholar API: Free academic search API with citation graphs, semantic similarity, TLDRs.

Connected Papers: Visual graph of related papers based on co-citation and co-reference.

Elicit: AI research assistant; extracts data from papers, finds relevant studies.

ResearchRabbit: Recommendation system for academic papers; exports to Zotero.

Claude with web search: Real-time paper discovery via arxiv + semantic search integration.

5. Building a Personal Research Feed

```
import feedparser
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
import numpy as np

# Fetch arxiv RSS for cs.AI + cs.LG
feeds = [
    feedparser.parse("https://arxiv.org/rss/cs.AI"),
    feedparser.parse("https://arxiv.org/rss/cs.LG"),
]
papers = [(e.title, e.summary) for f in feeds for e in f.entries]

# Your interest profile (titles of papers you liked)
liked = ["Attention Is All You Need", "Constitutional AI", "RAG"]

# TF-IDF similarity
corpus = [f"{t} {s}" for t, s in papers] + liked
vec = TfidfVectorizer(max_features=50000, stop_words='english')
X = vec.fit_transform(corpus)
sims = cosine_similarity(X[-len(liked):], X[:len(papers)]).mean(axis=0)

# Top-10 recommendations
top_idx = np.argsort(-sims)[:10]
for i in top_idx:
    print(f"{sims[i]:.3f} | {papers[i][0]}")
```

6. Future Directions

- Embedding-based search (voyage-3, E5) replacing TF-IDF for richer semantic matching
- LLM-generated paper summaries and TLDR annotations at ingestion time
- Citation graph traversal for finding foundational and follow-up work
- Multi-modal search: find papers containing specific figure types or results tables
- Agentic literature review: autonomous agent scans, summarizes, and synthesizes 50+ papers